

Intrinsic Triple-Enzyme Mimetic Activity of V_6O_{13} Nanotextiles: Mechanism Investigation and Colorimetric and Fluorescent Detections

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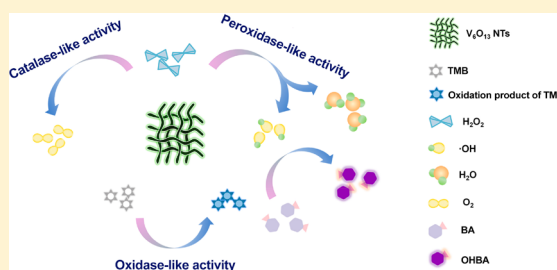
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Supporting Information

ABSTRACT: Nanozymes that display multiple activities have received increasing attention because they can be applied more widely than single-active nanozymes. In this paper, we demonstrated that the as-prepared V_6O_{13} nanotextiles (NTs) with interlaced network structure possess triple-enzyme mimetic activity for the first time. The intrinsic oxidase-like, peroxidase-like, and catalase-like catalytic behavior of V_6O_{13} NTs was investigated. Michaelis–Menten kinetics and high affinity for TMB and H_2O_2 was observed, indicating that V_6O_{13} NTs can be used as enzyme mimics. These multiactivity V_6O_{13} NTs were robust, inexpensive, and easily prepared. Based on the oxidase-like and peroxidase-like activity, a colorimetric biosensor for the glutathione (GSH) detection and a fluorescent system for the detection of H_2O_2 and glucose were established. The detection of GSH and glucose in health supplements and human serum samples was successfully applied through the proposed method.



1. INTRODUCTION

Nanostructured artificial enzymes (so-called nanozymes) are a new class of enzyme mimics.^{1,2} Owing to their remarkable advantages such as low cost, high stability, and good tunability in catalytic activities, nanozymes have attracted much attention.³ A host of works have verified that nanomaterials are able to mimic several classes of natural enzymes, such as peroxidase,⁴ catalase,⁵ oxidase,⁶ superoxide dismutase (SOD),^{7,8} and phosphotriesterase.⁹ Nanozymes can be classified into the following major categories: noble metals (e.g., Au,¹⁰ Ag,¹¹ Pt,¹² and Pd¹³), metal oxides (e.g., Fe_3O_4 ,¹⁴ Co_3O_4 ,¹⁵ V_2O_5 ,¹⁶ and CeO_2 ¹⁷), metal organic frameworks (e.g., ZIF-8,¹⁸ MIL-53(Fe),¹⁹ MIL-101(Fe),²⁰ and MIL-88²¹), carbon-based materials (e.g., carbon nanodots,²² graphene oxide,²³ and carbon nanotubes²⁴), and nanocomposites that consist of certain combinations of the above nanomaterials (e.g., MoS_2 –graphene oxide,²⁵ Co_3O_4 @ CeO_2 ,²⁶ Pt–Pd,²⁷ $g-C_3N_4$ – Fe_3O_4 ,²⁸ and hemin/ WS_2 ²⁹). Among the aforementioned nanozymes, multiactivity nanozymes have received tremendous attention. As the name implies, multiactivity nanozymes possess two or more catalytic activities, which could be utilized separately or integrally. For instance, nanoceria^{8,17,30} and Co_3O_4 ^{31,32} show three or four activities, each activity of these nanomaterials could be applied independently for the detection of different objects. The multiactivity of Prussian blue,³³ Au,³⁴ and CuO nanoparticles³⁵ could be used to establish self-cascade reaction systems to

determine different objects. These kinds of nanozymes integrate a multistep reaction into one single step so that the detection process becomes more convenient and efficient. Because of these outstanding properties of multiactivity nanozymes, much effort has been devoted to the development of multipurpose nanomaterials.

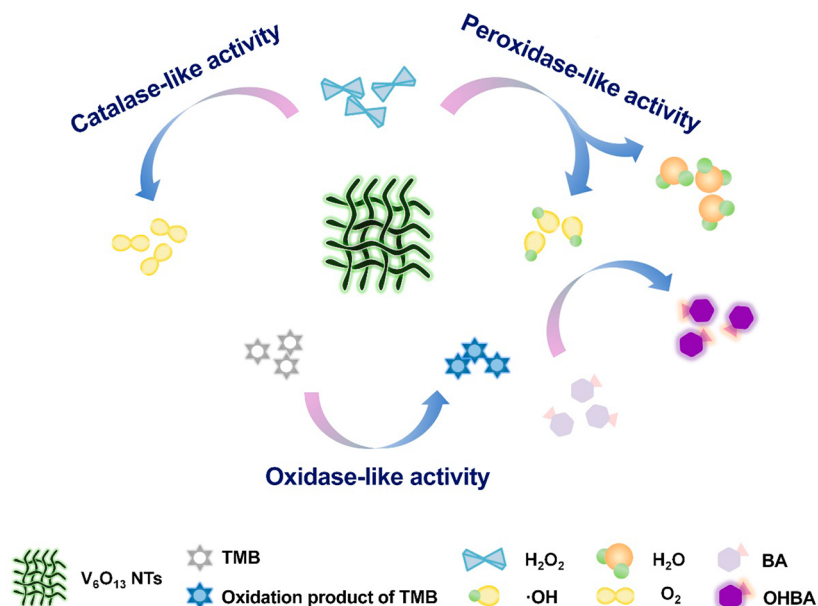
Vanadium oxides are a class of very important functional materials that have been used in a broad range of applications such as optoelectronic devices,³⁶ electrodes for lithium ion batteries,³⁷ gas sensors,³⁸ electrochemistry,³⁹ and catalysis.⁴⁰ However, among the many different compositions of vanadium oxide, only a few possess enzyme mimetic catalytic activity.^{16,41–43} Compared with the well-known V_2O_5 , mixed-valence V_6O_{13} is a relatively less-studied vanadium oxide. The presence of a coordination polyhedron gives V_6O_{13} an open structure that is capable of embedding organic groups and metal ions.⁴⁴ Because of its alternating single and double layer structure and share corners,^{45,46} V_6O_{13} has many more active sites that allow target objects (e.g., Li^+) to be inserted and extracted reversibly. Owing to the above characteristics, V_6O_{13} has been widely used in cathode materials for lithium-ion batteries⁴⁵ and supercapacitors⁴⁷ with high performance.

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Scheme 1. Schematic Representations of the Triple-Enzyme Mimetic Activity of V_6O_{13} NTs

Considering the superiority of the unique structure, we speculate that V_6O_{13} has certain catalytic properties that need to be investigated and elaborated.

In this work, we synthesized V_6O_{13} nanotextiles (NTs) with a unique interlaced network through a facile solution-redox-based self-assembly method, which favors fast electron/ion transport.⁴⁵ Then, the catalytic performance of the as-prepared V_6O_{13} NTs was studied. The result showed that V_6O_{13} NTs possessed oxidase-like, peroxidase-like, and catalase-like catalytic activity (Scheme 1). To the best of our knowledge, this is the first report of triple-enzyme mimetic activity of V_6O_{13} . The oxidase-like and peroxidase-like activity of V_6O_{13} NTs was studied in detail. Based on these two enzyme mimetic activities, a sensitive colorimetric biosensor was established for the detection of glutathione (GSH) and a fluorescence assay for the detection of H_2O_2 and glucose.

2. EXPERIMENTAL SECTION

2.1. Materials and Reagents. Potassium permanganate ($KMnO_4$) and ethyl acetate ($CH_3COOCH_2CH_3$) were supplied by Beijing Chemical Works (Beijing, China). 3,3',5,5'-Tetramethylbenzidine (TMB), 2,2'-azino-bis(3-ethylbenzothiazoline-6-sulfonic acid) (ABTS), *o*-phenylenediamine (OPD), and saccharides used in the current study were purchased from Alfa Aesar Co. Ltd. (Dora, FL, U.S.A.). Glucose oxidase (GOx) from *Aspergillus niger* was obtained from Sigma-Aldrich (St. Louis, MO, U.S.A.). Glucose and glutathione (reduced form, GSH) was purchased from TCI (Tokyo, Japan). Vanadium sulfate oxide hydrate ($VOSO_4 \cdot xH_2O$, 99.9%) was purchased from Alfa Aesar Co. Ltd. (Ward Hill, MA, U.S.A.). Hydrogen peroxide (H_2O_2 , 30%), benzoic acid (BA), sucrose, maltose, galactose, and fructose were supplied by Beijing Yili Chemical (Beijing, China). All chemicals were purchased as analytical grade and used as received without further purification. GSH chewable tablets and capsules were obtained from Cosway (M) Sdn Bhd and Swanson healthy products (Fargo, ND, U.S.A.), respectively.

2.2. Apparatus. Absorption spectra were recorded on a TU-1950 spectrophotometer (Beijing Purkinje General Instru-

ment Co. Ltd., Beijing, China). The fluorescence signals of the reaction systems were all recorded on an F-7000 fluorescence spectrophotometer (Hitachi, Japan). The morphological characteristics of the V_6O_{13} NTs were determined using a Hitachi S-4700 scanning electron microscope (SEM, Hitachi, Japan). X-ray diffraction (XRD) patterns were recorded on a D/MAX-2500 X-ray diffractometer (Rigaku, Tokyo, Japan) using $Cu K\alpha$ radiation ($\lambda = 1.54178 \text{ \AA}$) to study the crystal structure of the V_6O_{13} NTs. The valence states of the chemical bonds in the V_6O_{13} NTs were studied using an ESCALAB-250 X-ray photoelectron spectroscope (XPS, Thermo VG, Massachusetts, U.S.A.) using an $Al K\alpha$ radiation source. The Raman spectrum was obtained by a Renishaw inVia Reflex Laser Confocal Raman spectrometer with a 633 nm laser radiation source. O_2 generated during H_2O_2 decomposition was recorded using a HK-258 portable dissolved oxygen meter (Beijing HuaKe Instrument Power Equipment Research Institute, China).

2.3. Preparation of V_6O_{13} NTs. The V_6O_{13} NTs were prepared according to a method reported in the literature.⁴⁵ Briefly, the synthesis consisted of two steps. In the first step, an aqueous solution of $KMnO_4$ (0.02 M, 150 mL) was mixed with 40 mL of ethyl acetate and then refluxed at $85^\circ C$ until the pink color of the $KMnO_4$ solution disappeared; a brown suspension was obtained and separated from the upper ethyl acetate layer. In the second step, 0.06 M $VOSO_4$ solution was added into the resulting brown suspension under magnetic stirring at room temperature to generate a dark green suspension, which was continuously stirring for 2–4 h at room temperature. Finally, dark green products were collected by centrifugation and further washed with deionized water and absolute ethyl alcohol several times and dried at $40^\circ C$.

2.4. Kinetic Analysis. In the steady-state kinetic assays, catalytic reactions were carried out in a reaction volume of 5 mL HAc–NaAc buffer solution (50 mM, pH 4.5) under the optimized conditions with the various concentrations of one of the TMB or H_2O_2 substrates. The Michaelis–Menten constant was calculated using a Lineweaver–Burk plot:

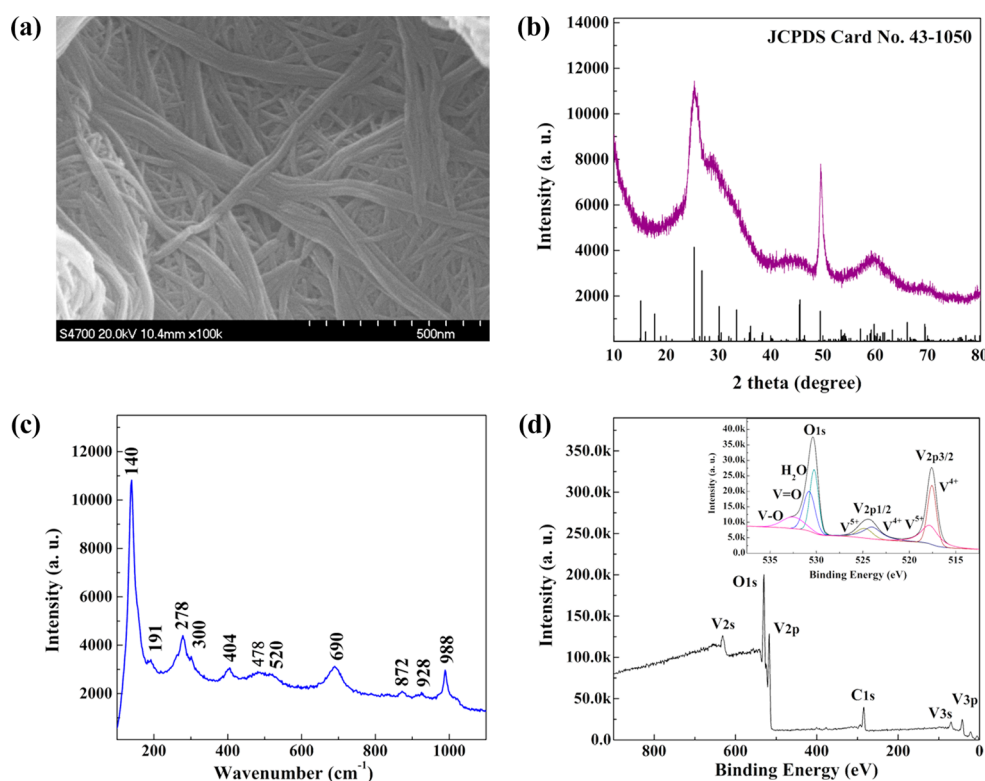


Figure 1. (a) SEM image, (b) XRD pattern, (c) Raman spectrum (excitation at 633 nm), and (d) XPS survey scan spectrum (inset: fitted curves for the HRXPS spectrum of V 2p and O 1s region) of as-prepared V_6O_{13} NTs.

$$\frac{1}{v} = \frac{1}{[S]} \frac{k_m}{v_{\max}} + \frac{1}{v_{\max}} \quad (1)$$

in which v is the initial velocity, $V_{\max}$ is the maximal reaction velocity, $[S]$ is the concentration of the substrate, and K_m represents the Michaelis–Menten constant, which is an indicator of the enzyme affinity to its substrate. The smaller the value of K_m , the stronger the affinity between the enzyme and substrate is.

2.5. Investigation of the Triple-Enzyme Mimetic Activity of the V_6O_{13} NTs. The oxidase-like activity of the V_6O_{13} NTs was studied by oxidizing TMB, ABTS, and OPD in the absence of H_2O_2 . In detail, 100 μL of V_6O_{13} NTs (1 mg mL^{-1}) and 300 μL of TMB (5 mM), ABTS (10 mM), or OPD (10 mM) were added into 4.6 mL of acetate buffer (50 mM, pH 3.0). The mixed solution was incubated at 30 $^{\circ}C$ for 20 min. Then, the corresponding color solution was transferred to a quartz cell for UV–vis detection at 652, 420, and 450 nm, corresponding to the catalytic product of TMB, ABTS, and OPD, respectively.

The peroxidase-like activity of V_6O_{13} NTs was studied by oxidizing TMB in the presence of H_2O_2 . In detail, 150 μL of V_6O_{13} NTs (0.5 mg mL^{-1}), 300 μL of TMB (5 mM), and various concentrations of H_2O_2 were added into the acetate buffer (50 mM, pH 4.5); the final reaction volume was 5 mL. The mixed solution was incubated at 30 $^{\circ}C$ for 20 min. Then, the corresponding color solution was transferred to a quartz cell for UV–vis detection at 652 nm.

The catalase-like catalytic activity of V_6O_{13} NTs was tested as follows: 5 mL of various concentrations of V_6O_{13} NTs and 5 mL of H_2O_2 (600 mM) were added into 20 mL of acetate buffer (50 mM, pH 4.5). A dissolved oxygen meter was adopted

to monitor the concentration of O_2 generated during H_2O_2 decomposition.

2.6. Applications of the Triple-Enzyme Activity of V_6O_{13} NTs.

2.6.1. Colorimetric Detection of GSH. GSH detection based on the oxidase-like catalytic activity of V_6O_{13} NTs was carried out according to a typical colorimetric analysis. In detail, 150 μL of V_6O_{13} NTs (0.5 mg mL^{-1}), 300 μL of TMB (5 mM), and a series volume of GSH (0.5 mM) were added into acetate buffer solution; the final reaction volume was 5 mL. The mixed solution was incubated at 30 $^{\circ}C$ for 20 min. Then, the corresponding color solution was transferred to a quartz cell for UV–vis detection at 652 nm.

2.6.2. Fluorescent Detection of H_2O_2 and Glucose. H_2O_2 detection based on the peroxidase-like catalytic activity of V_6O_{13} NTs was performed by using BA as a fluorescent substrate instead of TMB. The operating steps were as follows: 80 μL of the V_6O_{13} NTs (0.5 mg mL^{-1}), 500 μL of BA (14 mM), and 100 μL of H_2O_2 (8 mM) were added into 3.97 mL of phosphate buffered saline (PBS) solution (pH 4.5, 50 mM) and incubated at 30 $^{\circ}C$ for 10 min. After incubation, the pH of the reaction system was adjusted to 11.5 by using 330 μL of NaOH (1 M) to terminate the reaction and improve the fluorescence intensity. Then, the reaction system was transferred into a fluorescence cuvette for the detection of fluorescence. The fluorescence spectra of the reaction solutions were then recorded with a 298 nm excitation wavelength, and the fluorescence intensity at 405 nm was used for the quantitative analysis of H_2O_2 . A reaction system without H_2O_2 was regarded as a blank sample and ΔF was determined, whereby $\Delta F = F(H_2O_2, 405 \text{ nm}) - F(\text{blank}, 405 \text{ nm})$.

For glucose analysis, 100 μL of GOx (1 mg mL^{-1}) and various concentrations of glucose were added into a PBS solution (50 mM, pH 7.0) and incubated at 37 $^{\circ}C$ for 30 min.

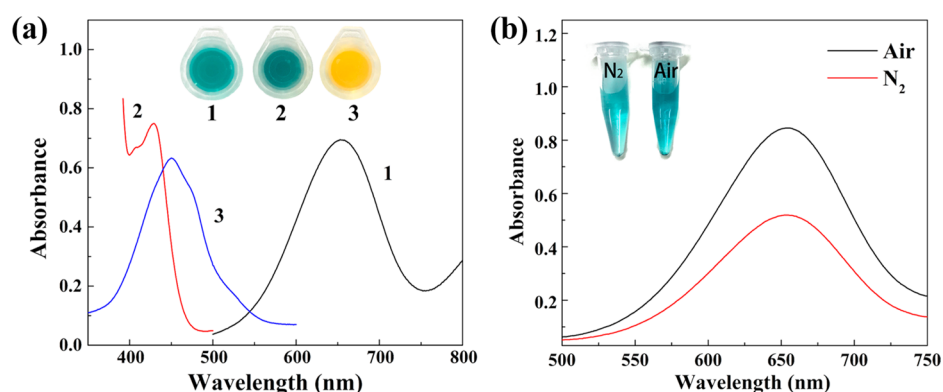


Figure 2. (a) V₆O₁₃ NTs catalyzing the oxidation of different chromogenic substrates to produce different colors in the absence of H₂O₂: (1) TMB, (2) ABTS, and (3) OPD. (b) Effect of O₂ on the direct oxidation of TMB by V₆O₁₃ NTs.

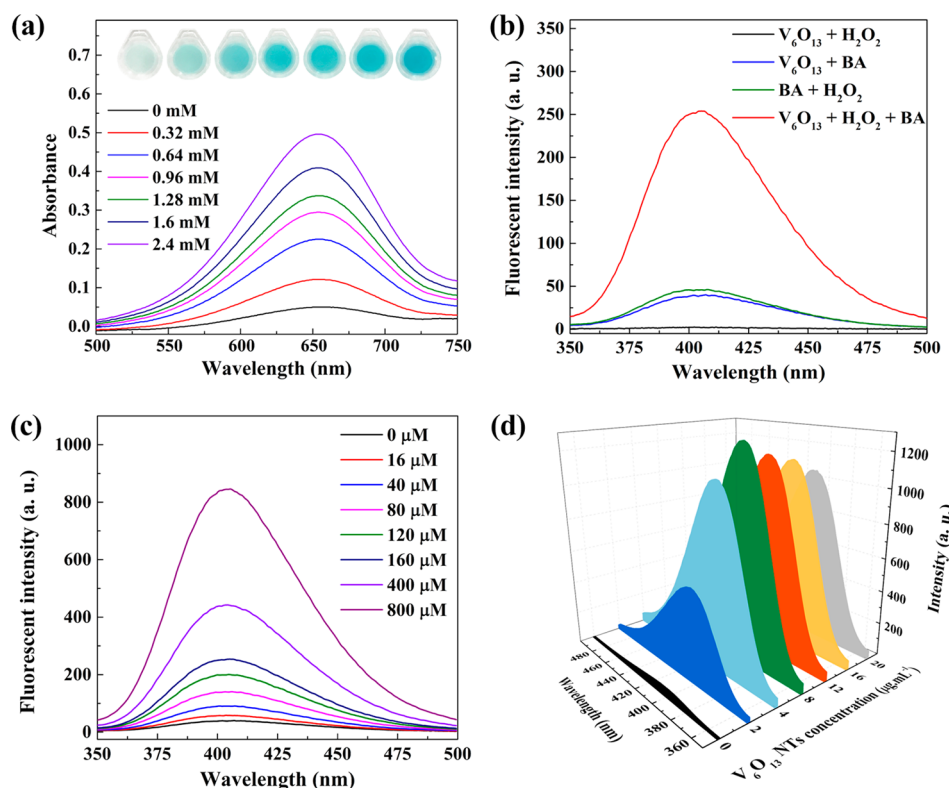


Figure 3. (a) V₆O₁₃ NTs catalyzing the oxidation of TMB in the presence of different concentrations of H₂O₂. (b) Fluorescence spectra of different reaction systems. (c) Effect of the concentration of H₂O₂ on the fluorescent reaction systems. (d) Effect of the concentration of V₆O₁₃ NTs on the fluorescent reaction systems.

Then, 100 μL of HCl (1 M) was added to adjust the pH of the reaction solution to 4.5; subsequently, 80 μL of the V₆O₁₃ NTs (0.5 mg mL⁻¹) and 500 μL of BA (14 mM) were added and incubated for another 10 min at 30 °C. Afterward, 330 μL of NaOH (1 M) was added, and the overall reaction system was transferred for fluorescence detection. To study the specificity of the proposed V₆O₁₃/H₂O₂/BA system toward glucose, 50 mM galactose, fructose, lactose, and maltose were used instead of 1 mM glucose. Similarly, we determined that $\Delta F = F(\text{glucose}, 405 \text{ nm}) - F(\text{blank}, 405 \text{ nm})$.

3. RESULTS AND DISCUSSION

3.1. Characterization of V₆O₁₃ NTs. The surface morphology of the V₆O₁₃ NTs was investigated by SEM analysis. As shown in Figure 1a, a layered stack interlaced

network structure of V₆O₁₃ NTs was observed. The crystallinity of the V₆O₁₃ NTs was determined by XRD analysis. As presented in Figure 1b, the diffraction pattern was well indexed to the monoclinic V₆O₁₃ phase (JCPDS no.: 43-1050). The Raman spectrum is depicted in Figure 1c. A strong peak located at 140 cm⁻¹ corresponded to B_g symmetry. The peaks located at 278, 404, and 990 cm⁻¹; 300 and 520 cm⁻¹; and 478 and 690 cm⁻¹ were assigned to the bending vibrations and stretch modes of O–V bonds, three-coordinated O–V(3) bonds, and bridging V–O–V, respectively.⁴⁸ This result further confirmed the formation of V₆O₁₃ and the absence of characteristic V₂O₅ and VO₂ modes. To determine the state of vanadium in the as-prepared V₆O₁₃ NTs, XPS measurements were performed, as shown in Figure 1d. The as-prepared products were composed of V, O, and C. C was mainly from the organic raw materials

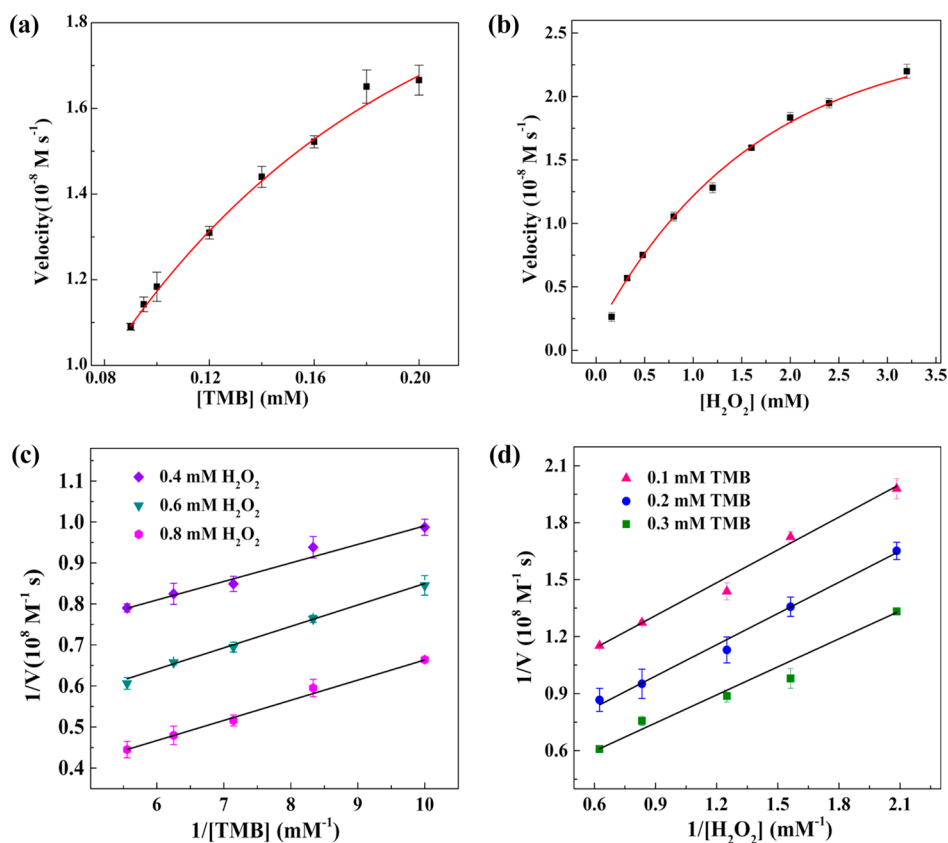


Figure 4. Steady-state kinetic assays of V_6O_{13} NTs. (a) The concentration of H_2O_2 was 0.6 mM and the TMB concentration was varied. (b) The concentration of TMB was 0.3 mM and the H_2O_2 concentration was varied. (c, d) The linear fitting of Lineweaver–Burk double reciprocal plots of the Michaelis–Menten equation, with the concentration of one substrate fixed and the other varied.

that remained in the final samples. The XPS spectra of V 2p and O 1s are displayed in the inset in Figure 1d. The bonding energies at 517.6 and 524.4 eV were assigned to V 2p_{3/2} and V 2p_{1/2}, which decomposed at 517.0, 517.8 eV and 524.0, 524.9 eV corresponding to V^{4+} (517.0, 524.0 eV) and V^{5+} (517.8, 524.9 eV).⁴⁹ This result validated the presence of mixed-valence vanadium oxides. The morphology of V_6O_{13} NTs after catalysis was also investigated. The SEM image (Figure S1) showed that V_6O_{13} NTs almost retain the interlaced network structure after catalysis, indicating that this structure is robust enough to sustain the catalysis behavior of V_6O_{13} NTs.

3.2. Triple-Enzyme Catalytic Activity of V_6O_{13} NTs. To investigate the triple-enzyme catalytic activity of V_6O_{13} NTs systematically, their oxidase-like, peroxidase-like, and catalase-like activities were investigated in detail.

3.2.1. Oxidase-Like Catalytic Activity of V_6O_{13} NTs. V_6O_{13} NTs could directly oxidize TMB, ABTS, and OPD into different color products in the absence of H_2O_2 (Figure 2a), which indicated the oxidase-like catalytic activity of V_6O_{13} NTs. To further study the oxidation of a chromogenic substrate such as TMB by V_6O_{13} NTs, the oxidizing agent (i.e., dissolved oxygen) in the reaction system was evaluated. The reaction vessel was degassed and then filled with N_2 . The V_6O_{13} NTs were dispersed in a buffer solution, and the dispersion was bubbled with N_2 for 30 min. The results are presented in Figure 2b; the color of the reaction system was lighter blue when bubbled with N_2 compared with that of the untreated system. Therefore, the dissolved O_2 played an important role as the electron acceptor in the oxidation of TMB. Similar to a natural enzyme and other nanozymes, the catalytic activity of V_6O_{13}

NTs depended on the temperature, pH, and concentrations of V_6O_{13} NTs and TMB. To optimize the reaction conditions, a series of experiments were carried out (Figure S2). As a result, the optimal reaction conditions were determined to be 15 $\mu\text{g mL}^{-1}$ of V_6O_{13} NTs, 0.3 mM of TMB, acetate buffer pH 3.0, at 30 °C incubated for 20 min. Subsequent experiments were carried out under the optimal conditions.

3.2.2. Peroxidase-Like Catalytic Activity of V_6O_{13} NTs. The peroxidase-like catalytic activity of V_6O_{13} NTs was investigated by catalyzing the oxidation of TMB in the presence of H_2O_2 . As shown in Figure 3a, the introduction of H_2O_2 gave rise to an increase in the absorbance at 652 nm. Furthermore, the absorbance of the reaction system increased with increasing concentration of H_2O_2 . These phenomena demonstrated that V_6O_{13} NTs possessed peroxidase-like catalytic activity.

The peroxidase-like catalysis mechanism of V_6O_{13} NTs was assumed to involve the generation of hydroxyl radicals ($\bullet\text{OH}$) during the reactions. To verify this hypothesis, benzoic acid (BA) was used as a fluorescent substrate. BA can react with $\bullet\text{OH}$ to generate a fluorescent product (OHBA), which has a characteristic emission peak at 405 nm when excited at 298 nm.^{50,51} As shown in Figure 3b, there was a strong characteristic emission peak at 405 nm in the system consisting of V_6O_{13} , H_2O_2 , and BA, whereas the fluorescence was weak in the system without H_2O_2 , V_6O_{13} , or BA. Then, we evaluated the fluorescence of different systems that contained different concentration of H_2O_2 . The results showed that the fluorescence of the reaction system increased with increasing H_2O_2 concentration, which indicated that the amount of

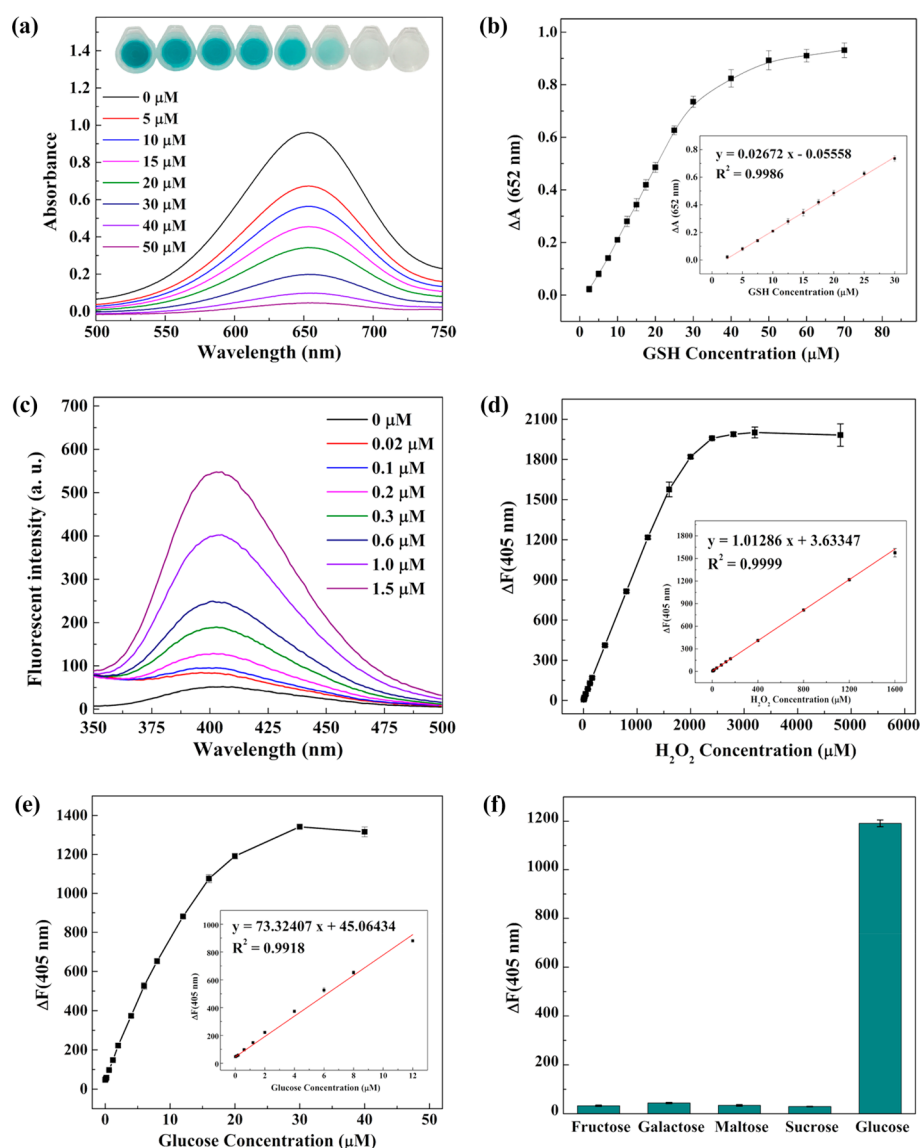


Figure 5. (a) UV-vis spectra of the V_6O_{13} -TMB system in the presence of varying concentrations of GSH. (b) Calibration plot for GSH determination. (c) Fluorescence spectra of V_6O_{13} -BA system in the presence of varying concentrations of glucose. (d) Calibration plot for H_2O_2 determination. (e) Calibration plot for glucose determination. (f) Fluorescence intensity of V_6O_{13} -BA system in the presence of other saccharides.

produced $\bullet OH$ was increased (Figure 3c). Therefore, we could confirm that V_6O_{13} NTs catalyzed H_2O_2 to produce $\bullet OH$.

Furthermore, we found that the concentration of the V_6O_{13} NTs had an effect on the fluorescence of the reaction systems. As shown in Figure 3d, when the concentration of the V_6O_{13} NTs increased from 2 to 8 $\mu g mL^{-1}$, the fluorescence intensity of the reaction system increased. As the concentration of the V_6O_{13} NTs continued to increase above 8 $\mu g mL^{-1}$, the fluorescence intensity of the reaction system decreased. This kind of activity has been reported in some other nanomaterials.^{52,53} We supposed that a chain reaction similar to that observed for MFe_2O_4 ($M = Mg, Ni, \text{ and } Cu$)⁵⁴ was present during the peroxidase-like behavior of V_6O_{13} NTs. When the concentration of the V_6O_{13} NTs was lower than 8 $\mu g mL^{-1}$, the reaction system produced an excess of $\bullet OH$, which resulted in an increase in the fluorescence intensity. Moreover, when the concentration of the V_6O_{13} NTs was higher than 8 $\mu g mL^{-1}$, the reaction system consumed more $\bullet OH$ to generate O_2 and therefore the amount of residual $\bullet OH$ reduced. This phenomenon also supported the theory that V_6O_{13} NTs

possessed catalase-like activity. In conclusion, the as-prepared V_6O_{13} NTs possessed peroxidase-like catalytic activity, and this catalytic activity originated from the formation of $\bullet OH$.

The peroxidase-like activity of V_6O_{13} NTs depended on the temperature, pH and concentrations of V_6O_{13} NTs and BA similar to horseradish peroxidase (HRP) and other nanozymes. To achieve the best catalytic behavior, a series of experiments were carried out to optimizing the reaction condition (Figure S3). As a result, the optimal reaction conditions were 8 $\mu g mL^{-1}$ of V_6O_{13} NTs, 1.4 mM of BA, PBS buffer pH 4.5, incubated at 30 $^{\circ}C$ for 10 min.

3.2.3. Catalase-Like Catalytic Activity of V_6O_{13} NTs. To verify the catalase-like activity of V_6O_{13} NTs, the dissolved oxygen concentration of the system consisting of V_6O_{13} and H_2O_2 was monitored by using a portable dissolved oxygen meter. The value of the dissolved oxygen concentration was recorded every 10 s for 15 min. As shown in Figure S4, the concentration of dissolved oxygen increased with the increase of the concentration of V_6O_{13} NTs from 0 to 0.6 $mg mL^{-1}$. This phenomenon indicated that V_6O_{13} NTs can decompose

Table 1. Analytical Results of the Determination of GSH in Real Samples

samples	reference value ^a (μM)	proposed method ^b (μM)	added (μM)	found (μM)	recovery (%)	RSD($n = 3$) (%)
chewable tablets	9.8	9.9	5.0	14.8	99.2	3.6
			10.0	20.3	104.4	2.9
			15.0	25.3	102.6	1.7
capsules	14.4	14.3	5.0	19.3	97.2	1.8
			10.0	24.4	99.9	1.4
			15.0	29.3	99.2	2.9

^aResults given in the specifications. ^bResults obtained by this work.

Table 2. Determination of Glucose in Human Serum Samples ($n = 3$)

sample ^a	glucose concentration (mmol L^{-1})		glucose added (mmol L^{-1})	glucose found (mmol L^{-1})	recovery (%)	RSD (%)
	hospital method ^b	proposed method ^c				
1	5.0	5.04	10.00	15.25	102.10	2.64
			20.00	24.53	97.45	1.29
2	5.4	5.34	10.00	15.11	97.69	1.10
			20.00	25.18	99.23	0.95

^aThe human serum samples were diluted 100-fold to ensure that the concentration of glucose in human serum samples was within the linear range of the calibration curve. ^bResults obtained by the School clinic of Beijing University of Chemical Technology. ^cResults obtained by this work.

H_2O_2 into O_2 and provided strong proof of the catalase-like activity of V_6O_{13} NTs.

3.3. Steady-State Kinetic Analysis of V_6O_{13} NTs. To further study the catalytic mechanism, the steady-state kinetic parameters for TMB and H_2O_2 were determined and compared with HRP and other vanadium oxide peroxidase-like nanozymes. The typical Michaelis–Menten curves (Figure 4a,b) and Lineweaver–Burk plots (Figure 4c,d) were obtained and the important kinetic parameters such as K_m and $V_{\max}$ were derived. The K_m and $V_{\max}$ value for V_6O_{13} NTs toward TMB was 0.153 mM and $2.99 \times 10^{-8} \text{ M s}^{-1}$ and toward H_2O_2 was 1.51 mM and $3.12 \times 10^{-8} \text{ M s}^{-1}$. Remarkably, the K_m value of V_6O_{13} NTs with TMB and H_2O_2 was lower than that of HRP, suggesting that V_6O_{13} NTs had a higher affinity toward TMB and H_2O_2 than toward HRP. The high performance of the V_6O_{13} NTs may be attributed to the presence of more “active sites” on the surface of the V_6O_{13} nanoarchitecture than HRP,^{16,55} which has only one Fe ion at the active center.⁵⁶ Compared with other vanadium oxide nanozymes (Table S1), the K_m value with TMB of V_6O_{13} NTs was close to $\text{VO}_2(\text{A})$ and $\text{VO}_2(\text{B})$ owing to the analogous structure. The K_m value with H_2O_2 as the substrate was higher than that for $\text{VO}_2(\text{A})$, which indicated that a higher H_2O_2 concentration was required to achieve maximal activity for V_6O_{13} NTs.¹⁵ Figure 4c,d showed the double reciprocal plots of initial velocity against the concentration of one substrate in a certain range of concentrations of the second substrate. The characteristic parallel lines revealed a ping-pong mechanism, implying that the V_6O_{13} NTs reacted with the first substrate and then released the first product before reacting with the second substrate.

3.4. Applications of Triple-Enzyme Activity of V_6O_{13} NTs. **3.4.1. Colorimetric Detection of Glutathione.** Because the oxidase-like activity of nanozymes can be used for antioxidant detection,⁵⁷ a V_6O_{13} –TMB biosensor system was established for glutathione (GSH) detection. As shown in Figure 5a, as the concentration of GSH was increased, the reaction system exhibited a paler color. Under the optimal reaction condition, the calibration curve for the detection of GSH was obtained. The calibration data of ΔA_{652} ($\Delta A_{652} = A_{(\text{blank})} - A_{(\text{GSH})}$) versus GSH concentration are shown in Figure 5b. The results indicated that the limit of detection was

$0.63 \mu\text{M}$ ($S/N = 3$) with a linear range from 2.5 to $30 \mu\text{M}$. To investigate the selectivity of the proposed method, colorimetric responses of the proposed detection system were investigated in the presence of some potential interferences (common amino acids, saccharides, and cations) that probably coexist with GSH in health care products. The results shown in Figure S5 revealed that the potential interferences have no significant effect on the V_6O_{13} –TMB biosensor system even their concentrations were 100-fold higher than GSH. Then, the established GSH detection system was applied to two real samples (chewable tablets and capsules) under the optimal condition. As shown in Table 1, the concentrations of GSH detected in the two samples using the proposed method were very close to the standard values given in the specifications. The recoveries of GSH from the samples were in a range from 97.2 to 104.4%, and the RSD values were in range from 1.4 to 3.6%. Therefore, the proposed method could prove to be feasible for the practical application of GSH detection.

3.4.2. Fluorescent Detection of H_2O_2 and Glucose. Based on the peroxidase-like activity of V_6O_{13} NTs, a V_6O_{13} –BA fluorescent system for H_2O_2 and glucose determination was established. Figure 5c revealed that the fluorescence intensity of the system increased as the glucose concentration increased. In a certain concentration range, the value of ΔF_{405} ($\Delta F_{405} = F(\text{H}_2\text{O}_2/\text{glucose}) - F(\text{blank})$) exhibited a good linear correlation to the concentration of $\text{H}_2\text{O}_2/\text{glucose}$. Under the optimal condition, the standard curve for the detection of H_2O_2 and glucose was plotted. As shown in Figure 5d, ΔF_{405} showed an excellent linear relationship with the concentration of H_2O_2 in a broad range of 8.0–1600.0 μM . The linear regression equation was $\Delta F = 1.0129C (\mu\text{M}) + 3.6335$ ($R^2 = 0.9999$) with a detection limit of 6.41 μM . For glucose detection (Figure 5e), ΔF_{405} was proportional to the concentration of glucose in the range of 0.2–12 μM , with a detection limit of 0.02 μM , which was more sensitive than that reported in the most of previous reports (Table S2). The linear regression equation was $\Delta F = 73.3241C (\mu\text{M}) + 45.0643$ ($R^2 = 0.9918$). This glucose detection system showed excellent selectivity against other sugars (fructose, lactose, maltose, and sucrose). As depicted in Figure 5f, even if the concentrations of the interfering substances were as high as 50 mM, there was no significant

fluorescence responses in the reaction systems. Hence, it can be further applied to the detection of glucose in a real sample. To further investigate the feasibility of the established method for detecting glucose in biological samples, glucose in human serum samples was determined. The obtained results are listed in Table 2. This proposed method was in good agreement with the results obtained by a hospital method, and the recoveries for glucose in human serum samples were in the range of 97.45%–102.1%, suggesting that this V_6O_{13} –BA system can be applied for the determination of glucose in medical diagnosis and biological analysis.

4. CONCLUSION

In summary, V_6O_{13} NTs were synthesized by a facile solution-redox-based self-assembly method and exhibited triple-enzyme mimetic activity: oxidase-like, peroxidase-like, and catalase-like activity. The as-prepared V_6O_{13} NTs had a unique interlaced network structure and favored fast electron/ion transport. These characteristics were conducive to good catalytic performance of V_6O_{13} NTs. Based on the prominent oxidase-/peroxidase-like activity, colorimetric and fluorescent biosensors were established for the detection of GSH, H_2O_2 , and glucose. The developed methods have the advantage of high sensitivity, easy operation, and good practicability. Furthermore, the developed methods were applied for the detection of GSH in health care products and the determination of glucose in human serum samples. The methods displayed satisfactory results with high sensitivity and accuracy. We believe that the V_6O_{13} NTs hold great promise for environmental monitoring, biological analysis, and clinical diagnosis applications in the future.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge on the ACS Publications website at DOI: 10.1021/acs.iecr.7b04821.

SEM image of V_6O_{13} NTs after catalysis. The optimization of reaction conditions of the oxidase-like activity of V_6O_{13} NTs. The optimization of reaction conditions of the peroxidase-like activity of V_6O_{13} NTs. Effect of the V_6O_{13} NTs concentration on the O_2 generation from H_2O_2 decomposition. Colorimetric responses of V_6O_{13} –TMB system in the presence of potential interferences. Comparison of apparent kinetic parameters among natural enzymes and nanozymes. Comparison of peroxidase-like activity in the linear range and detection limit for glucose detection between V_6O_{13} NTs and other nanozymes. (PDF)

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Notes

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